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4 / 4 submitted

Level 2 — 25 marks/question**L2-Q1. subsequence (M)**

c

CODE

```
#include <stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int a[n],lis[n],max=0;
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
        lis[i]=1;
    }
    for(int i=0;i<n;i++){
        for(int j=0;j<i;j++){
            if(a[i]>a[j]&&lis[i]<lis[j]+1){
                lis[i]=lis[j]+1;
            }
        }
    }
    for(int i=0;i<n;i++){
        if(lis[i]>max)
            max=lis[i];
    }
    printf("Length of the longest increasing subsequence:%d\n",max);
    return 0;
}
```

INPUT

```
8
10 22 9 33 21 50 41 60
```

OUTPUT

```
Length of the longest increasing subsequence:5
```

C

CODE

```
#include<stdio.h>

struct product{
    char name[50];
    float order;
    int Customer;
    float payment;
};

int main(){
    struct product p;
    printf("Welcome to the Online Shopping System!\nMenu: \n1.Add Product to Cart\n2.Place order\n3.Make Payment\nEnter your C
hoice: \n");
    printf("\nAvailable Products: \n1.Laptop($800)\n2.Smartphone($500)\nEnter product number to add Cart: ");

}
```

OUTPUT

```
Welcome to the Online Shopping System!
Menu:
1.Add Product to Cart
2.Place order
3.Make Payment
Enter your Choice:

Available Products:
1.Laptop($800)
2.Smartphone($500)
Enter product number to add Cart:
```

L2-Q3. rotation (M)

C

CODE

```
#include <stdio.h>
#include <string.h>
int main(){
    char s1[100],s2[100],temp[200];
    scanf("%99s",&s1);
    printf("Enter the first string:%s",s1);
    scanf("%99s",&s2);
    printf("\nEnter the second string:%s",s2);
    if(strlen(s1)==strlen(s2)){
        strcpy(temp,s1);
        strcat(temp,s1);
        if(strstr(temp,s2)!=NULL){
            printf("\nThe second string is a rotation of the first string.\n");
            return 0;
        }
    }
}
```

INPUT

```
ABCD
CDAB
```

OUTPUT

```
Enter the first string:ABCD
Enter the second string:CDAB
The second string is a rotation of the first string.
```

L2-Q4. symmetric (M)

C

CODE

```
#include <stdio.h>
int main(){
    int rows,cols;
    scanf("%d",&rows);
    printf("Enter number of rows:%d",rows);
    scanf("%d",&cols);
    printf("Enter number of columns:%d",cols);
    int matrix[rows][cols];
    for(int i=0;i<rows;i++){
        for(int j=0;j<cols;j++){
            scanf("%d",&matrix[i][j]);
        }
    }
    if(rows!=cols){
        printf("\nThe matrix is not symmetric.\n");
        return 0;
    }
    for(int i=0;i<rows;i++){
        for(int j=0;j<cols;j++){
            if(matrix[i][j]!=matrix[j][i]){
                printf("\nThe matrix is not symmetric.\n");
                return 0;
            }
        }
    }
    printf("\nThe matrix is symmetric.\n");
    return 0;
}
```

INPUT

```
3
3
1 2 3
2 4 5
3 5 6
```

OUTPUT

```
Enter number of rows:3Enter number of columns:3
The matrix is symmetric.
```

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0 / 4 submitted

Level 2 — 25 marks/question

L2-Q1. even and odd (M)

— Not Submitted —

L2-Q2. Training (M)

— Not Submitted —

L2-Q3. permutations (M)

— Not Submitted —

L2-Q4. identity matrix (M)

— Not Submitted —